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Branch and Bound Method

This method is applicable to both pure and mixed IPP.

Let the given LPP will be maximize $Z = Cx$.

S to c: $AX \leq b$

$x \geq 0$ and integers

In the optimal solution sum of the variables say x_r is not an integer, then $x_r^* < x_r < x_r^* + 1$ where x_r^* and $x_r^* + 1$ are consecutive non-negative integers.

This x_r must satisfy one of the two conditions

$$x_r \leq x_r^* \quad \text{or} \quad x_r \geq x_r^*$$

Now we found two diff sub problems

~~Sub probr~~

Sub-problem 1:

Max $Z = Cx$

$AX \leq b$

$x \geq 0$

$x_r \leq x_r^*$

Sub problem 2

Max $Z = Cx$

$Ax \leq b$

$x \geq 0$

$x_r \geq x_r^* + 1$

Thus we have branched the original problem into two subproblems

The branching proces eliminates that portion of the feasible region that contains no feasible integer solution.

If any subproblem yields an optimum integer solution it is not further branched.

Use branch and bound technique to solve the following

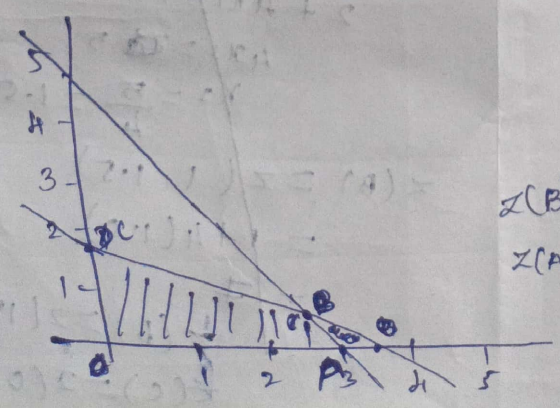
Max $Z = x_1 + 4x_2$.

S to c $2x_1 + 4x_2 \leq 7$

$5x_1 + 3x_2 \leq 15$ $\quad x_1, x_2 \geq 0$ & integers.

I: $x_1 = 0 \Rightarrow x_2 = \frac{7}{4} = 1.75$ $\quad x_2 = 0 \Rightarrow x_1 = 7/2 = 3.5$ $\quad (1.75, 3.5)$

II: $x_1 = 0 \Rightarrow x_2 = 5$ $\quad x_2 = 0 \Rightarrow x_1 = 3$ $\quad (3, 5)$



I $2x_1 + 4x_2 = 7$

II $5x_1 + 3x_2 = 15$

I $\times 3 \Rightarrow 6x_1 + 12x_2 = 21$

II $\times 4 \Rightarrow 20x_1 + 12x_2 = 60$

$-14x_2 = -39$

$x_2 = \frac{-39}{-14}$

$Z(B) = 4.1$

$Z(A) = Z(3,0) = 3$

$Z(C) = Z(0, 1.7)$

$= 6.8$

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Max $Z = 68$ at $x_1 = 0$ $x_2 = \frac{7}{4} \Rightarrow 1 < x_2 \leq 2$

$x_2 \leq 1$ or $x_2 \geq 2$

SB1

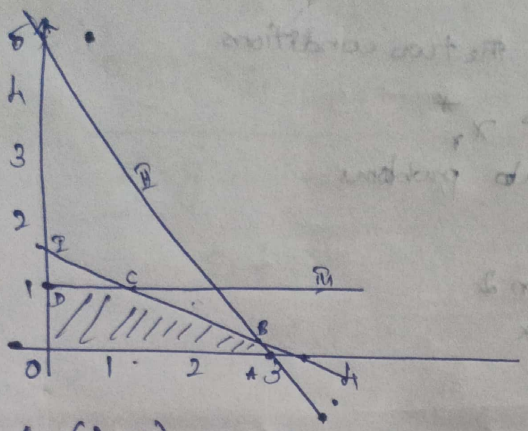
Max $Z = x_1 + 4x_2$

S to C

$2x_1 + 4x_2 \leq 7$

$5x_1 + 3x_2 \leq 15$

$x_2 \leq 1$



A (3, 0) $\Rightarrow Z(A) = 3$

B $Z(B) = 4.18$

C = I ∩ II $Z(D) = 4$

$2x_1 + 4x_2 = 7$

$2x_1 + 4 = 7$

$2x_1 = 3$

$x_1 = 3/2$

$Z(B) = Z(2.7, 0.5)$

E $(\frac{3}{2}, 1)$

$Z(C) = Z(\frac{3}{2}, 1) = \frac{3}{2} + 4 = \frac{11}{2}$

Max $Z = \frac{11}{2}$ at $x_1 = 3/2$ and $x_2 = 1$

$1 \leq x_1 \leq 2$ ($x_1 \leq 1$ / $x_1 \geq 2$)

Sub prob 3

Max $Z = x_1 + 4x_2$

S to C:

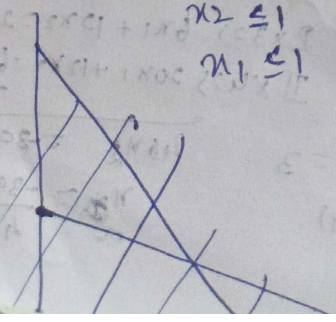
$2x_1 + 4x_2 \leq 7$

$5x_1 + 3x_2 \leq 15$

~~$x_1 \leq 1$~~

$x_2 \leq 1$

$x_1 \leq 1$



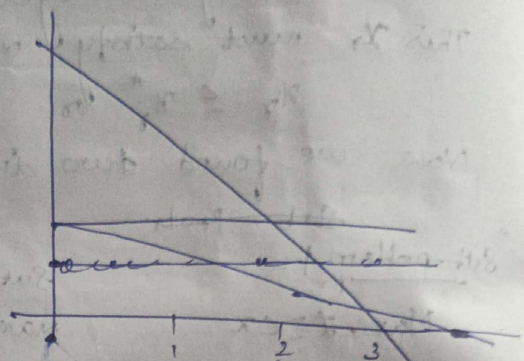
SB2

Max $Z = x_1 + 4x_2$

$2x_1 + 4x_2 \leq 7$

$5x_1 + 3x_2 \leq 15$

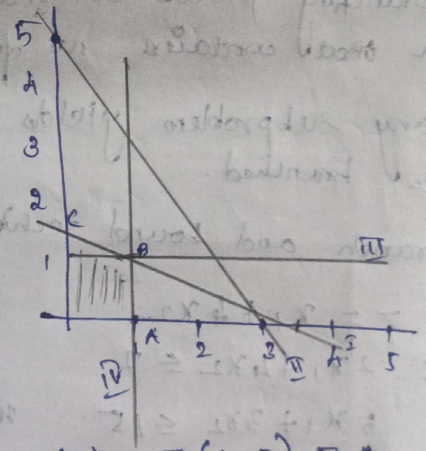
$x_2 \geq 2$



No feasible solution

Fathomed

~~Sub prob 4~~



$Z(A) = Z(1, 0) = 1$

~~$Z(C) = Z(1, 1.7)$~~ 0, 1.7

$2x_1 + 4x_2 = 7$

$2 + 4x_2 = 7$

$4x_2 = 5$

$x_2 = \frac{5}{4} = 1.5$

$Z(B) = Z(1, 1.5)$

$= 1 + 4(1.5)$

$= 7$

$Z(B) = 2(1, 1) = 5$

$Z(C) = 2(0, 1) = 4$

2
1.5x4
6.0

~~Sub prob 3~~

Max $z = 5$

at $x_1 = 1, x_2 = 1$

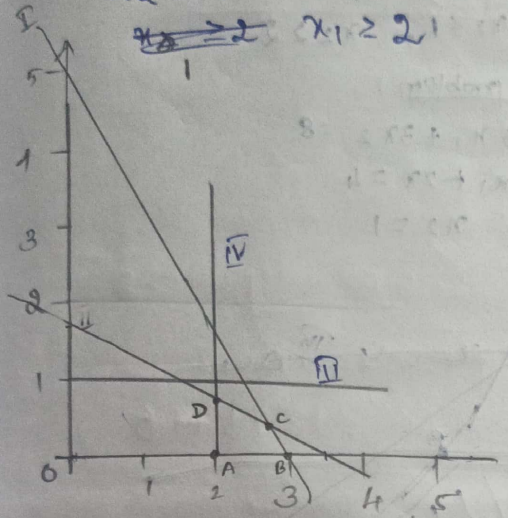
Sub problem 4 $\boxed{z = x_1 + 4x_2}$

$2x_1 + 4x_2 \leq 7$

$5x_1 + 3x_2 \leq 15$

$x_2 \leq 1$

~~$x_1 \geq 2$~~ $x_1 \geq 2$



$A(2,0)$ $z(A) = 2$

$B(3,0)$ $z(B) = 3$

$C = 2$ and 1 $z(C) = 4 \cdot 18$

$C = 2 \cdot 7, 0 \cdot 35$

~~D~~ $4x_2 = 3$

$x_2 = 3/4$

$D = (2, 3/4)$

$z(D) = 5$

Max $z =$ ~~4.18~~ 5

~~at $x_1 = 2 \cdot 7$ and $x_2 = 0 \cdot 35$~~

at $x_1 = 2$, $x_2 = 3/4 = 0 \cdot 75$

$0 \leq x_2 \leq 1$

$x_2 \leq 0$ $x_2 \geq 1$

Sub problem - 5

Max $z = x_1 + 4x_2$

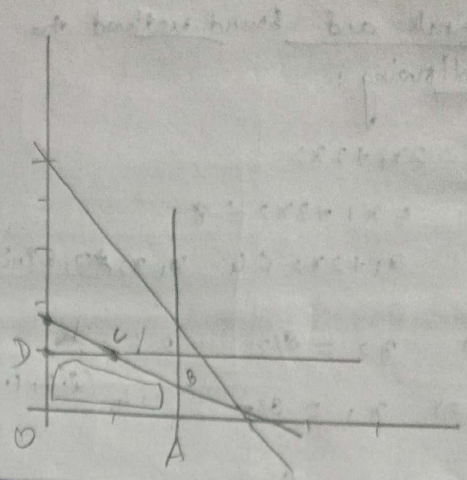
$2x_1 + 4x_2 \leq 7$

$5x_1 + 3x_2 \leq 15$

$x_2 \leq 1$

$x_1 \geq 2$

$x_2 \leq 0$ No feasible



Sub problem 6

$z = x_1 + 4x_2$

$2x_1 + 4x_2 \leq 7$

$5x_1 + 3x_2 \leq 15$

$x_2 \leq 1$

$x_1 \geq 2$

$x_2 \geq 1$

no solution, Infeasible

$A(0,1)$ $C(0, 1 \cdot 7)$

$B(1,1)$

$z(A) = 4$

$z(B) = 5$

$z(C) =$

Max $z = 5$ at $x_1 = 1$

$x_2 = 1$

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Use branch and bound method to solve following:

Max $Z = 2x_1 + 2x_2$

S to c $5x_1 + 3x_2 \leq 8$

$x_1 + 2x_2 \leq 4$ $x_1, x_2 \geq 0$, and integer

I

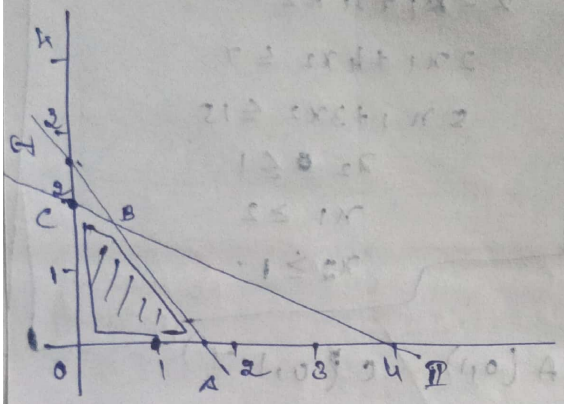
$x_1 = 0 \Rightarrow x_2 = 8/3 = 2.6$

$x_2 = 0 \Rightarrow x_1 = 8/5 = 1.6$ (2.6, 1.6)

II

$x_1 = 0 \Rightarrow x_2 = 2$

$x_2 = 0 \Rightarrow x_1 = 4$ (4, 0)



A (0 1.6, 0)

C ~~(4, 0, 0) (2, 0, 0)~~ C(0, 2)

$5x_1 + 3x_2 = 8$

$(-)\ 5x_1 + 10x_2 = 20$

$-7x_2 = -12$

$x_2 = 12/7 = 1.7$

$x_1 + 2(1.7) = 4$

$x_1 + 3.4 = 4$

$x_1 = 0.6 = 4/7$

$Z(0) = 0$

$Z(A) = Z(0, 1.6) = 3.2$

$Z(B) = Z(4/7, 12/7) = 32/7 = 4.6$

$Z(C) = Z(0, 2) = 4$

Max $Z = 4.6$ at

$x_1 = 4/7$, $x_2 = 12/7$

The solution is optimal but not an integer.

We use branch and bound method to get integer values.

We choose 12/7

$1 \leq x_2 \leq 2$

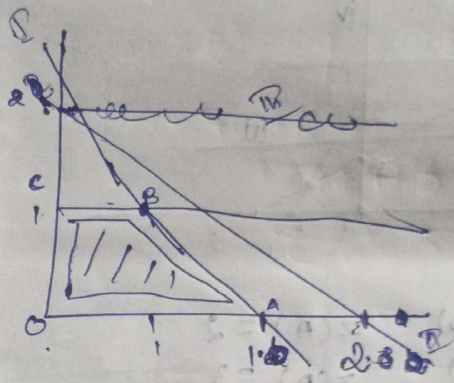
$x_2 \leq 1$, $x_2 \geq 2$

Sub problem 1

$5x_1 + 3x_2 = 8$

$x_1 + 2x = 4$

$x_2 = 1$



B(1, 1)

A (1.6, 0)

C (0, 1)

$Z(A) = Z(1.6, 0) = 3.2$

$Z(B) = Z(2, 1) = 4$

$Z(C) = (0, 1) = 2$

Max $Z = 4$ at $x_1 = 1, x_2 = 1$

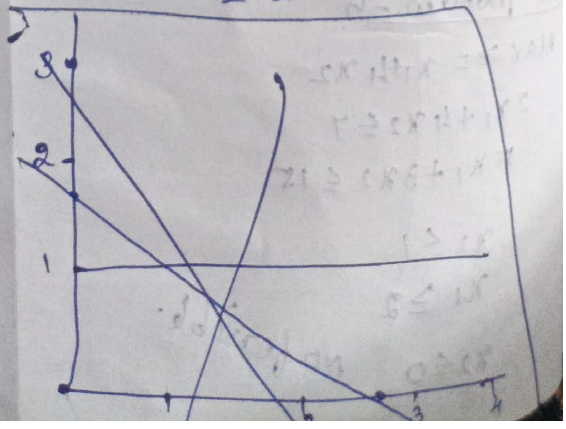
The solution is fathomed.

Sub problem 2

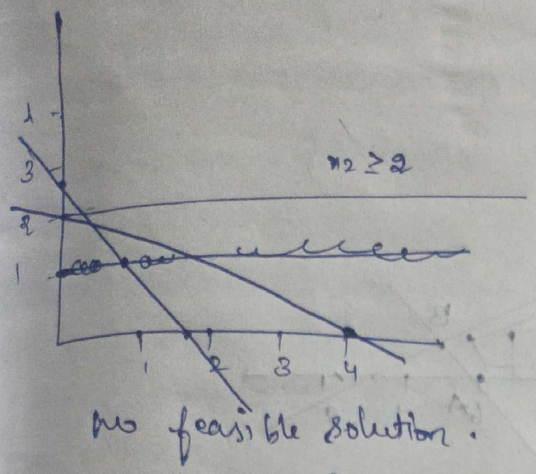
$5x_1 + 3x_2 = 8$

$x_1 + 2x_2 = 4$

$x_2 = 2$



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No feasible solution.

So the subproblem is fathomed

Choose $\frac{4}{7}$ $7\overline{)40}$

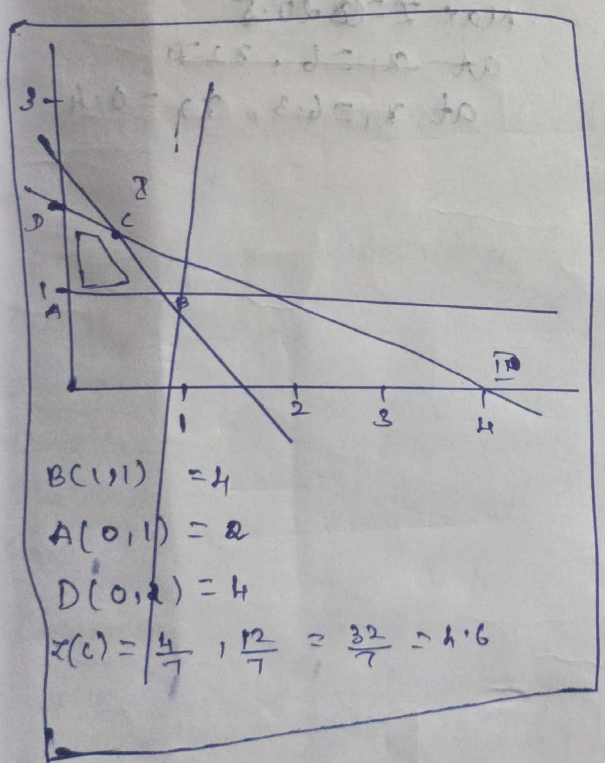
$0 \leq x_1 \leq 1$

$x_1 \leq 0$ $x_1 \geq 1$

$5x_1 + 3x_2 = 8$

$x_1 + 2x_2 = 4$

$x_1 = 1$

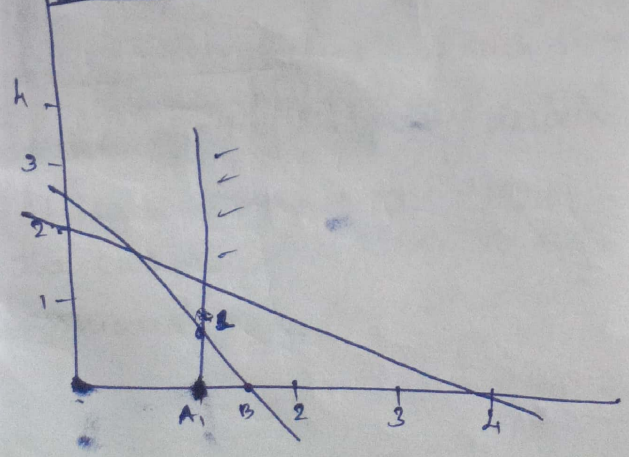


$B(1,1) = 4$

$A(0,1) = 2$

$D(0,2) = 4$

$z(C) = \frac{4}{7}, \frac{12}{7} = \frac{32}{7} = 4.6$



$z(A) = 2$

$A(1,0)$ ~~$z(A) = 2$~~

$B(1.6,0)$ $z(B) = 3.2$

~~$C(1,1)$~~

$5x_1 + 3x_2 = 8$

$C(1,1)$ $z(C) = 4$

Max $z = 4$

at $x_1 = 1$, $x_2 = 1$

Max $z = 3x_1 + 4x_2$

S to C

$7x_1 + 16x_2 \leq 52$

$x_1 - 2x_2 \leq 18$

D $x_1 = 0$ $x_2 = \frac{52}{16} = 3.25$

$x_2 = 0 \Rightarrow x_1 = \frac{52}{7} = 7.4$

A $x_2 = 0 \Rightarrow x_2 = -9$

$x_2 = 0 \Rightarrow 18/5 = x_1$

0, 3.25

C $x_1 = 0 \Rightarrow x_2 = \frac{52}{16} = 3.25$

7.42, 0

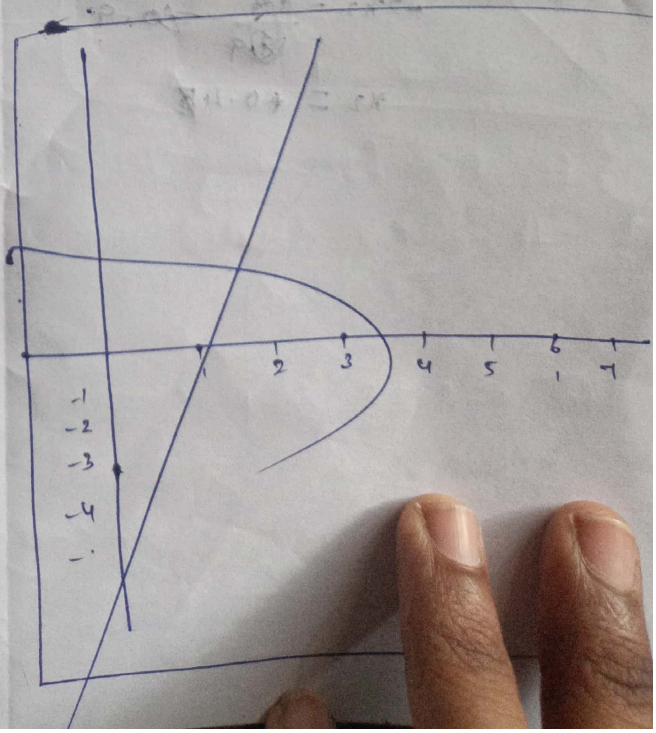
$x_2 = 0 \Rightarrow x_1 = 6$

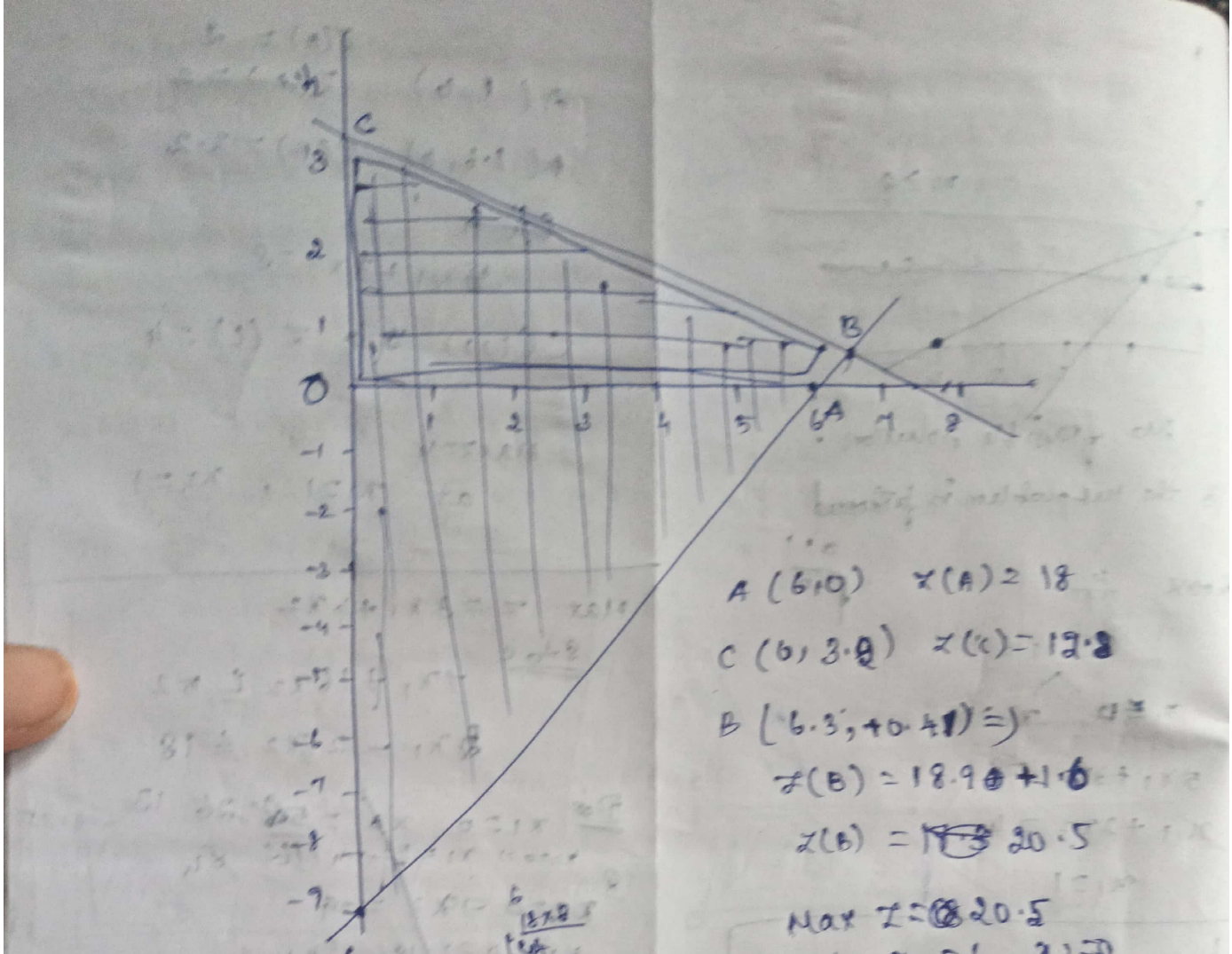
$x_2 = 0 \Rightarrow x_1 = \frac{52}{7} = 7.42$

0, -9

B $x_1 = 0 \Rightarrow x_2 = -9$

E $x_2 = 0 \Rightarrow x_1 = 6$ (6, 0)





$x_1 + 16x_2 = 52$

$24x_1 - 16x_2 = 144$

$31x_1 = 196$

$x_1 = \frac{196}{31} = 6.3$

$3(6.3) + 16x_2$

$3x_1 - 2x_2 = 18.$

$18.9 - 2x_2 = 18$

$-2x_2 = 18 - 18.9 = -0.9$

$x_2 = +0.4$

A (6, 0) $z(A) = 18$

C (6, 3.0) $z(C) = 12.3$

B (6.3, +0.4) $\Rightarrow$

$z(B) = 18.9 + 1.6$

$z(B) = 20.5$

Max $z = 20.5$

at $x_1 = 6.3, x_2 = 0.4$

Ur 3

(i) The dynamic programming technique decomposes the original problem in n variables into n sub problems or n stages each in one variable.

(ii) The solution is obtained in an orderly manner by starting from 1 stage to the next and is completed after the final stage is reached

Need of dynamic programming

It is required that

* All the decission of the combinations are specified
* The optimal policy can be selected only after all the combinations are evaluated.

(iii) Bellman's principle of optimality

An optimal policy has the property that whatever be the initial state and initial decisions, the remaining decisions must constitute the optimal policy for the state resulting from the first decision

Characteristi DPP

* The problem can be divided into stages, with a policy decission required at each stage.
* The effect of the policy decision at each stage is to transform the current state into a state associated with the next state

Applications of DPP

* In the production area, this technique has been used for production, scheduling and Employement smoothening problems.
* It has been used to determine the inventory level and for formulating the inventory recording
* It is used to determine the optimal combination of advertising media.

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pure and mixed differe
mixed integer program. -2(g).
is ~~from equation to~~

24/7/26

Divide a positive quantity c into n parts in which a way that their product is maximum. (or).

Q.#

Maximize $z = y_1 \cdot y_2 \cdots y_n$

S.t.o c

$y_1 + y_2 + \cdots + y_n = c$

$y_i \geq 0 \; ; \; i = 1, 2, 3, \ldots n$

Sol

To develop recursive equation

Let y_i be the ith part of c and each i may be regarded as a stage.

The optimal solution at each stage or obtained by usual classical method.

Let $f_n(c)$ be the maximum attainable product, $y_1 + y_2 + \cdots + y_n = c$

c is divided into n parts $y_1 \cdot y_2 \cdots y_n$.

Then $f_n(c)$ becomes a function of n. ~~for n=1 (one stage problem~~

$f_n(c) = y_1 \cdot y_2 \cdots y_n$

for n=1 (one stage problem)

$f_1(c) = c$ trivial

$\max_{0<x<c} f_1(c) = c$

for n=2 (two stage problem)

Let $y_1 = x_1, \; y_2 = c - x$

For n=1 (one stage problem)

Here c is divided into one part only that is $y_1 = c$

$\therefore \; f_1(c) = c$

For n=2 (2 stage problem)

Here c is divided into two parts $y_1 = x, \; y_2 = c - x$ the

$y_1 + y_2 = x + c - x = c$

$f_2(c) = \max_{0 \leq x \leq c} \{ y_1 \cdot y_2 \} = \max_{0 \leq x \leq c} \{ x \cdot c - x \}$

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$$= \max_{0 \le x \le c} \{ x \cdot f_1(c-x) \}$$

$n=3$ (Three stage problem)

Here c is divided into three parts y_1, y_2, y_3

let $y_1 = x$, $y_2 + y_3 = c - x$

$$y_1 + y_2 + y_3 = x$$

$c-x$ is further divided into two parts whose maximum attainable product $y_2 \cdot y_3$ is $f_2(c-x)$. then

$$f_3(c) = \max_{0 \le x \le c} \{ y_1 y_2 y_3 \}$$

$$= \max_{0 \le x \le c} \{ x \; f_2(c-x) \}$$

In general the recursive equation for the n stage problem is $f_n(c) = \max_{0 \le x \le c} \{ x \cdot f_{n-1}(c-x) \}$

To solve the recursive equation

for $n=2 \Rightarrow f_2(c) = \max \{ x \cdot f_1(c-x) \}$

$$\max \{ x \cdot (c-x) \}$$

The function $x \cdot (c-x)$ will be maximum if $\frac{d}{dx}[x(c-x)] = 0$

$$\frac{d}{dx}[x(c-x)] = 0$$

$$x(-1) + 1(c-x) = 0$$

$$-2x + c = 0$$

$$x = c/2$$

$$f_2(c) = \max \left\{ \frac{c}{2} \cdot c - \frac{c}{2} \right\} = \frac{c}{2}\left(\frac{c}{2}\right) = \left(\frac{c}{2}\right)^2$$

optimal policy $(op) = \left\{ \frac{c}{2}, \frac{c}{2} \right\}$ and $f_2(c) = \left(\frac{c}{2}\right)^2$

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for $n=3$

$$f_3(c) = \max\{x f_2(c-x)\}$$

$$= \max\left\{x\cdot\left(\frac{c-x}{2}\right)^2\right\}$$

The function $x\cdot\left(\frac{c-x}{2}\right)^2$ will be maximum if

$$\frac{d}{dx}\left[x\left(\frac{c-x}{2}\right)^2\right]=0$$

$$\frac{1}{4}\left[x\cdot 2(c-x)(-1)+1(c-x)^2\right]=0$$

$$(c-x)\left[-2x+c-x\right]=0$$

$$(c-x)\left[c-3x\right]=0$$

$c-x=0$	$c-3x=0$
$c=x$	$c=3x$
	$x=c/3$

$$f_3(c) = \max\left\{x\left(\frac{c-x}{2}\right)^2\right\}$$

$$= \max\left\{\frac{c}{3}\left(\frac{(c-\frac{c}{3})}{2}\right)^2\right\} = \frac{c}{3}\left(\frac{2c}{6}\right)^2 = \left(\frac{c}{3}\right)^3$$

Let us assume that the optimal policy for $n=m$ is

$\left(\frac{c}{m}, \frac{c}{m}\cdots\frac{c}{m}\right)$ & $f_m(c) = \left(\frac{c}{m}\right)^m$

Now for $n=m+1$

$$f_{m+1}(c) = \max_{0} \max\{x f_m(c-x)\}$$

$$= \max\left\{x\left(\frac{c-x}{m}\right)^m\right\}$$

The function $x\cdot\left(\frac{c-x}{m}\right)^m$ attain its maximum when

$\frac{d}{dx}\left[x\left(\frac{c-x}{m}\right)^m\right)=0$ we get $x=\frac{c}{m+1}$

$f_{(m+1)}c = \left(\frac{c}{m+1}\right)^{m+1}$ ∴ The result is true for $n=m+1$

∴ Hence by mathematical induction, the optimal policy is

$\left\{\frac{c}{n}, \frac{c}{n}, \frac{c}{n}\cdots\frac{c}{n}\right\}$ & $f_n(c) = \left(\frac{c}{n}\right)^n$.

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a) Solve the following by DP.

Minimize $z = y_1 + y_2 + \cdots + y_n$

S to C

$y_1 \cdot y_2 \cdots y_n = b$ and $y_1, y_2 \cdots y_n \geq 0$ (or)

factorize a positive quantity b into n factors in such a way so that their sum is a minimum.

Sol To develop a recursive equation

Let $f_n(b)$ will be the minimum attainable sum $y_1 + y_2 \cdots + y_n$ when the positive quantity b is factorized into n factors $y_1 \cdot y_2 \cdots y_n$.

For n = 1 (1 stage problem):

Here b is factorized into one factor.

$$y_1 = b$$

$$\therefore f_1(b) = b \text{ (trivial)}$$

For n = 2 (2 stage problem):

Here b is divided into two factors.

$$y_1 = x \;,\; y_2 = \frac{b}{x}$$

so that $y_1 \cdot y_2 = b$

then $f_2(b) = \max_{0 \leq x \leq b} \{ y_1 + y_2 \}$

$$= \max_{0 \leq x \leq b} \left\{ x + \frac{b}{x} \right\} = \max \left\{ x + f_1\left(\frac{b}{x}\right) \right\}$$

For n = 3 three stage problem:

b is factorized into three factors.

$$y_1 = x \quad y_2 y_3 = \frac{b}{x}$$

so that $y_1 \cdot y_2 \cdot y_3 = b$

then $f_3(b) = \max \{ y_1 + y_2 + y_3 \}$

$$= \max \left\{ x + f_2\left(\frac{b}{x}\right) \right\}$$

In general the recursive equation for the n stage problem is

$$f_n(b) = \max \left\{ x + f_{n-1}\left(\frac{b}{x}\right) \right\}$$

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Solving recursive equation:

for $n=2$ we have $f_2(b) = \max\{x + f_1(\frac{b}{x})\}$.

$$f_2(b) = \max\{x + \frac{b}{x}\}$$

The function $x + \frac{b}{x}$ will get maximum if $\frac{d}{dx}(x+\frac{b}{x}) = 0$

$$f_2(b) = \sqrt{b} + \frac{b}{\sqrt{b}} = \frac{b+b}{\sqrt{b}} = 2b^{1/2}$$

$o.p = \{b^{1/2}, b^{1/2}\}$, $f_2(b) = 2b^{1/2}$

for $n=3$

$$f_3(b) = \max\{x + f_2(b/x)\}$$

$$= \max\{x + 2(b/x)^{1/2}\}$$

$$\frac{d}{dx}\left[x + 2\frac{\sqrt{b}}{\sqrt{x}}\right] = 0$$

$$1 + 2\sqrt{b}\left(\frac{-1}{2}\right)x^{-3/2} = 0$$

$$\frac{1 - \sqrt{b}}{x^{3/2}} = 0$$

$$x^{3/2} - \sqrt{b} = 0$$

$$x^{3/2} = b^{1/2}$$

$$x^3 = b$$

$$x = b^{1/3}$$

$$f_3(b) = b^{1/3} + 2\left(\frac{b}{b^{1/3}}\right)^{1/2}$$

$$= b^{1/3} + 2b^{1/2}\, b^{(-1/3)1/2}$$

$$= b^{1/3} + 2b^{1/2}\, b^{-1/6}$$

$$= b^{1/3} + 2^{4/12}$$

$$= b^{1/3} + 2b^{1/3} = 3b^{1/3}$$

$\frac{1}{2} - \frac{1}{6} \quad \frac{6-2}{12} = \frac{4}{12}$

$\frac{-1}{3}$

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Solution of LPP by DPP

Solve the following LPP using DP principles

Max $Z = 2x_1 + 5x_2$

S.to $2x_1 + x_2 \leq 43$

$2x_2 \leq 46$, and $x_1, x_2 \geq 0$

So, The problem consist of two decission variables and two constrains.

Here the problem has two stages and 2 state variables.

Let (B_{1j}, B_{2j}) be the state of the system at stage 1 & 2.

$f_j(B_{1j}, B_{2j})$ be the optimal value of the objective function for stage 1 & 2. using backward computational procedure, we have

~~$f_j(B_{1j}, B_{2j})$~~

$$f_2(B_{12}, B_{22}) = \max_{\text{constrain } \{0 \leq x_2 \leq B_{12},\ 0 \leq x_2 \leq B_{22}\}} \{5x_2\}$$

$$= 5 \max_{\substack{0 \leq x_2 \leq B_{12} \\ 0 \leq x_2 \leq B_{22}/2}} \{x_2\} \rightarrow$$

since $\max(x_2)$ which satisfies $0 \leq x_2 \leq B12$, $0 \leq x_2 \leq B_{22}/2$ is the minimum of $B_{12}, B_{22}/2$

$$x_2^{\#} = \max\{x_2\} = \min\{B_{12}, B_{22}\}$$

$$f_2(B_{12}, B_{22}) = 5\min\left\{B_{12}, \frac{B_{22}}{2}\right\} \rightarrow (2)$$

$$f_1(B_{11}, B_{21}) = \max_{0 \leq 2x_1 \leq B_{11}} \{2x_1 + f_2(B_{11} - 2x_1, B_{21} - 0)\}$$

$$f_2(B_{11} - 2x_1, B_{21}) = 5\min\left\{B_{11} - 2x_1, \frac{B_{21}}{2}\right\}$$

$$f_1(B_{11}, B_{21}) = \max_{0 \leq x_1 \leq \frac{B_{11}}{2}} \left\{2x_1 + 5\left(\min B_{11} - 2x_1, \frac{B_{21}}{2}\right)\right\}$$

$B_{11} = 43$ $B_{21} = 46$

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$$f_1(B_{11}, B_{21}) = \max_{0 \le x_1 \le \frac{B_{11}}{2}} \left\{ 2x_1 + 5 \min\left(43 - 2x_1, \frac{46}{2}\right) \right\}$$

Rough $\quad 43 - 2x_1 = 23 \Rightarrow x_1 = 10$

$$\min(43 - 2x_1, 23) = \begin{cases} 23, & \text{if } 0 \le x_1 \le 10 \\ 43 - 2x_1, & \text{if } 10 \le x_1 \le 23 \end{cases}$$

$$f_1(B_{11}, B_{21}) = \max_{0 \le x_1 \le \frac{43}{2}} \left\{ 2x_1 + 5 \begin{cases} 23 & \text{if } 0 \le x_1 \le 10 \\ 43 - 2x_1 & \text{if } 10 \le x_1 \le 23 \end{cases} \right\}$$

$$= \max_{0 \le x_1 \le \frac{43}{2}} \begin{cases} 2x_1 + 115 & \text{if } 0 \le x_1 \le 10 \\ 2x_1 + 215 - 10x_1 & \text{if } 10 \le x_1 \le 23 \end{cases}$$

$\Rightarrow$ $f = \max$ of $\{2(10) + 15, 215 - 8(10)\}$

$= 135$ at $x_1 = 10$ and $x_2 = 23$

$$x = \min\left\{ B_{11} - 2x_1, \frac{B_{21}}{2} \right\}$$

$$= \min\left\{ 43 - 2(10), \frac{46}{2} \right\} \Rightarrow x_2 = 23$$